Unit Seven

Scalar Control of AC Motors

Electrical methods are used to regulate(control) the speed of induction motors by affecting the stator circuit or the rotor circuit of the motor.

The process of regulating the speed of the AC motors is done by influencing the stator circuit by controlling the scalar value of the stator circuit parameters, namely the stator voltage and the magnitude of stator frequency, using different types of power converters according to the speed regulation method.

Controlling Stator Voltage and Stator Frequency Techniques

The voltage and frequency values are changed using specific laws that depend on the type of industrial machinery used and the type of power converter.

In the following, we will show the cases used for stator voltage and stator frequency variations.

The stator instantaneous electromotive force is:

$$e_1 = -W\frac{d\phi}{dt}$$

Where, W :is the stator windings number of turns.

Instantaneous magnetic flux value:

$$\phi = \phi_{max} \sin \omega t$$

Where, (ω): is the velocity of change of the sine wave and is equal to:

$$\omega = 2\pi f_1$$

We find by substitution and derivation the following:

$$e_1 = 2\pi f_1 \phi_{max} W \cos(\omega t)$$

We find the rms value of the stator electromotive force:

$$E_1 = 4.44 \phi_{max} f_1 W$$

Stator voltage vector value using ac motor phasor diagram:

$$\dot{V}_1 = -\dot{E}_1 + \dot{I}_1 \dot{Z}_1$$

We write the rms value of the stator voltage by neglecting the voltage drop on the stator circuit:

$$\mathbf{V_1 \approx E_1}$$

Substituting the stator emf value and simplifying we get the relationship between stator voltage and stator frequency:

$$\boldsymbol{V_1 = K\phi f_1}$$

From the previous relationship there are several possibilities for speed control of ac motor by varying stator voltage and stator frequency :

1- Varying stator voltage value and maintain the constant stator frequency.

$$V_1 = Var \qquad f_1 = f_{1N} = const.$$

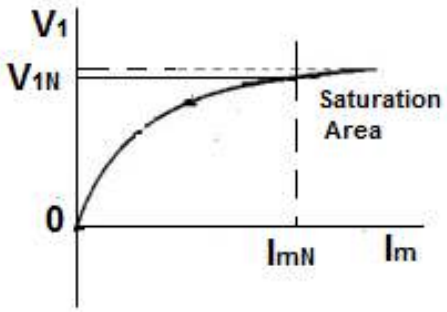
We get the following relationship:

$$\mathbf{V_1 \propto \phi}$$

That is, the motor magnetic flux is proportional to the stator voltage.

The increase in voltage leads to an increase in the magnetic flux, and the transition to work on the saturation zone on the magnetizing / Hysteresis characteristic, as shown in the figure below.

This case variation causes motor windings over current and over heating.



Hysterical characteristic of the electric motor

The decrease in voltage causes the magnetic flux to decrease, and thus the electromagnetic torque of the motor. In the case that the torque of the load remains constant, it causes the motor to be overloaded (the increase in current in the motor windings) and the increase of windings temperature.

The ac motor electromagnetic torque relationship shows that in case of constant load torque when decreasing motor flux, the rotor current increases to perform motor electromagnetic torque equals to load torque as a result the motor overheated.

$$T = K\Phi\ I_2^{/} \sin\varphi_2^{/} = cons\tan t$$

Therefore, the voltage must be changed with constant frequency while keeping the motor loading capacity constant.

$$V_1 \le V_{1N} \qquad f_1 = f_{1N} = const.$$

In this case, an AC Voltage Regulator (Controller) is used to control industrial machinery with fan load characteristic.

2- Varying stator frequency and maintaining the stator voltage constant.

$$f_1 = Var \qquad V_1 = V_{1N} = const.$$

We get the following relationship:

$$\phi\ f_1 = const.$$

$$\phi \propto \frac{1}{f_1}$$

That is, the flux is inversely proportional to the frequency. The decrease in the frequency leads to an increase in the magnetic flux and work in the saturation region, and this is not permissible to protect the motor from overheating, and therefore the stator frequency must be increased with constant stator voltage and maintaining constant motor loading capacity.

$$f_1 \ge f_{1N} \qquad V_1 = V_{1N} = const.$$

In this case, a Frequency Converter is used to control the speed of constant power industrial machinery (production machinery).

3- Varying stator voltage and stator frequency together

By using an inverter, the voltage and frequency can be changed together according to laws depending on the type of industrial machinery used.

Stator field angular speed is:

$$\omega_o = \frac{2\pi f_1}{P/2}$$

Any variation in frequency will cause a change in (ω_o). So, the motor speed regulation is done in two directions, above and below the natural characteristic.

Variable Voltage and Variable Frequency (VVVF) Laws

These laws are based on the condition that the motor loading capacity remains constant (constant motor loading capacity) for all stator voltage and stator frequency values.

$$\lambda = \text{const.}$$

$$\lambda = \frac{T_{max}}{T_L} = \frac{3V_1^2}{2\omega_o(R_1 + \sqrt{R_1^2 + X_{SC}^2})T_L} = \text{Const.}$$

Where:

$$X_{SC} = 2\pi f_1 L_{SC}$$

$$\omega_o = \frac{2\pi f_1}{P/2}$$

Assuming that $R_1 \cong 0$, i.e. by neglecting the stator voltage drop, we find the following:

$$\lambda = \frac{3V_1^2}{2\frac{(2\pi)}{P/2} f_1 (2\pi f_1 L_{S.C})T_L} = \text{const.}$$

By simplifying the previous relationship, we get:

$$\lambda = A\frac{V_1^2}{f_1^2 T_L} = const.$$

We write the previous relationship for the nominal parameters values:

$$A\frac{V_1^2}{f_1^2 T_L} = A\frac{V_{1N}^2}{f_{1N}^2 T_{LN}}$$

$$\frac{V_1^2}{f_1^2} = \frac{V_{1N}^2}{f_{1N}^2}\frac{T_L}{T_{LN}}$$

The general law relationship of stator voltage and stator frequency variation is:

$$\frac{V_1}{f_1} = \frac{V_{1N}}{f_{1N}}\sqrt{\frac{T_L}{T_{LN}}}$$

It is worth noting that the control of stator voltage and stator frequency is proportional and decreasing.

$$V_1 \leq V_{1N} \text{ and } f_1 \leq f_{1N}$$

The Industrial Machineries Variable Stator Voltage and Stator Frequency Laws

The first case: Transporting Machinery, which is constant load torque machinery.

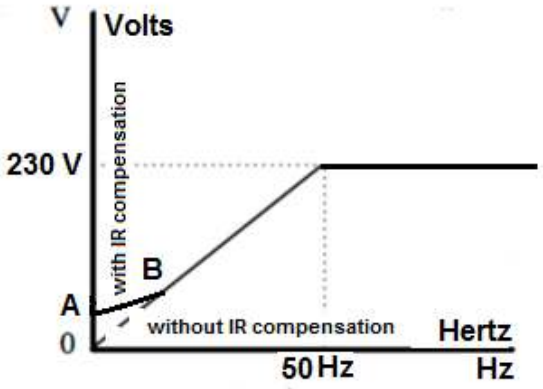
$$T_L = T_{LN}$$

And by applying this case to the general law of stator voltage and stator frequency variation, we get:

$$\frac{V_1}{f_1} = \frac{V_{1N}}{f_{1N}} = const.$$

$$\frac{V_1}{f_1} = \text{const.}$$

As shown in the figure below:



At lower frequencies the relationship becomes non-linear (A-B).

This is to compensate the neglected stator voltage drop.

Figure (7-1) shows the torque-speed characteristics of transporting machinery for $\frac{V_1}{f_1} = \text{const.}$

The speed control direction in this case is below the natural characteristic, parallel characteristics are obtained with a constant load torque equal to the rated torque.

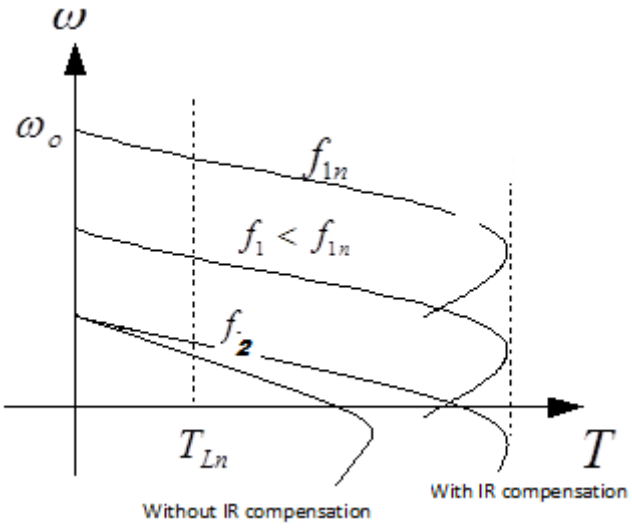


Figure (7-1) Torque-speed characteristics of transporting machinery

The law of VVVF is adopted by neglecting the voltage drop in the stator circuit IR = 0.

The value of the stator voltage when controlling the motor speed at the lower frequencies is small comparing to the voltage drop. This causes a decrease in the motor loading capacity at the fixed load torque due to the decrease in the maximum torque of the motor, which leads to the motor overheating.

To keep constant motor loading capacity and to avoid motor overloading and overheating, the stator voltage drop to be compensated by IR compensating as shown in figure (7-1).

The second case: Production Machinery, which is called constant load power machinery.

Load torque relationship:

$T_L\omega = T_{LN}\omega_N$ **, that means** $P_L = const.$

Constant load power .

The load torque is inversely proportional to the speed:

$$\frac{T_L}{T_{LN}} = \frac{\omega_N}{\omega}$$

Given that the angular speed is directly proportional to the frequency:

$$\frac{T_L}{T_{LN}} = \frac{f_{1N}}{f_1}$$

We substitute this relationship in VVVF general law, and we get:

$$\frac{V_1}{f_1} = \frac{V_{1N}}{f_{1N}} \sqrt{\frac{f_{1N}}{f_1}}$$

$$\frac{V_1}{\sqrt{f_1}} = \frac{V_{1N}}{\sqrt{f_{1N}}}$$

The production machinery VVVF law is:

$$\frac{\mathbf{V_1}}{\sqrt{\mathbf{f_1}}} = \mathbf{const.}$$

In this case, the speed regulation is with constant power, that is, the load torque is inversely proportional to the speed. And we get the regulated characteristics below the natural characteristic as in Figure (7-2).

The speed can be regulated with constant power above the natural characteristic as in Fig. (7-2) by increasing the stator frequency with the constant stator voltage.

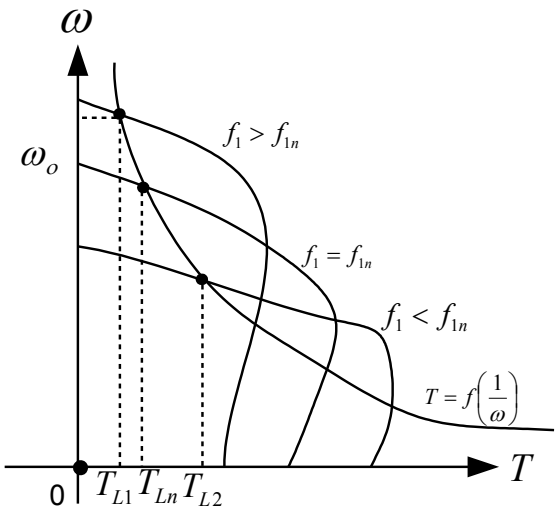


Figure (7-2) Production machinery torque-speed characteristics

The third case: Pumps, compressors and fan (ventilator), which have Fan Load Characteristic.

Load torque relationship:

$$T_L = T_{LN}\left(\frac{\omega}{\omega_N}\right)^2$$

The fan load characteristic is:

$$\mathbf{T_L \propto (\omega)^2}$$

We substitute this relationship in VVVF general law, and we get:

$$\frac{T_L}{T_{LN}} = \left(\frac{\omega}{\omega_N}\right)^2 = \left(\frac{f_1}{f_{1N}}\right)^2$$

$$\frac{V_1}{V_{1N}} = \frac{f_1}{f_{1N}} \cdot \frac{f_1}{f_{1N}} = \frac{f_1^2}{f_{1n}^2}$$

The fan load characteristic machinery VVVF law is:

$$\frac{\mathbf{V_1}}{\mathbf{f_1^2}} = \mathbf{const.}$$

The speed regulation is below the natural characteristic as shown in Figure (7-3).

Since the torque of the load is proportional to the square of motor speed $T_L \propto (\omega)^2$, the power is proportional to the speed cube $P_L \propto (\omega)^3$.

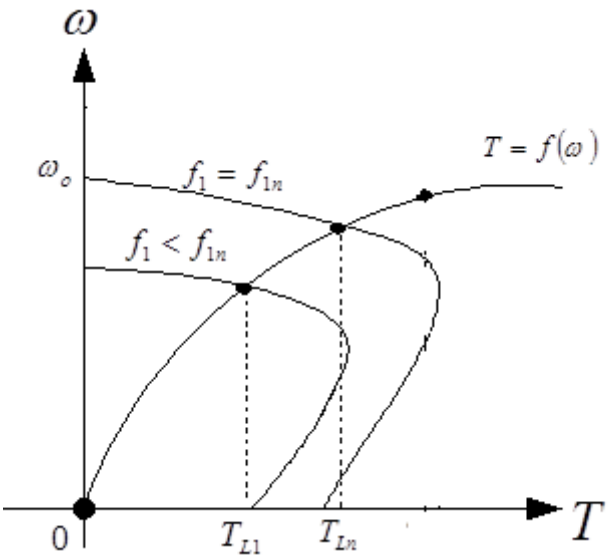
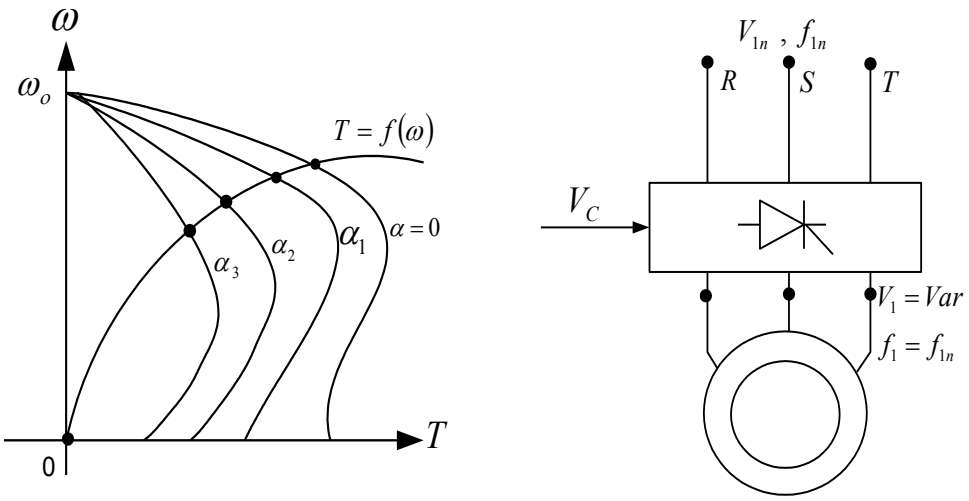


Figure (7-3) Torque-speed characteristics of fan load characteristic machinery

AC Voltage Controller Three-Phase AC Motor Speed Control Drive

Thyristors are used as shown in Figure (7-4-A) to regulate the stator voltage ,which is the controller output at a fixed frequency according to the following relationship:



B) Torque-speed characteristics **A) Power circuit of drive system**

Figure (7-4) AC Voltage Controller Three-Phase AC Motor Speed Control Drive

Stator voltage relationship is:

$$\mathbf{V_1 = V_{1n} \cos\alpha}$$

$$V_1 = f(\alpha)$$

Where:

α **: Thyristor firing angle.**

In this case, we control motor speed by varying the stator voltage with fixed frequency.

Speed control analysis

The requirement for speed control is constant motor loading capacity:

$$\lambda = \frac{\mathbf{T_{max}}}{\mathbf{T_L}} = \mathbf{Const.}$$

$$\frac{T_{\max(V_1)}}{T_L} = \frac{T_{\max(V_{1N})}}{T_{LN}}$$

$$T_{L1} = T_{LN} \frac{T_{\max(V_1)}}{T_{\max(V_{1N})}} = T_{LN} \left(\frac{V_1}{V_{1N}} \right)^2$$

The motor torque is proportional to the square of the stator voltage as follows:

$$\left(\mathbf{T_{max} \propto V_1^2} \right)$$

$$T_{L1} = T_{LN} \left(\frac{V_{1n} Cos\alpha}{V_{1N}} \right)^2 = T_{LN} Cos^2 \alpha$$

The speed control is below the natural characteristic and the regulation range is small. This system is used to regulate the speed of the fan load characteristic machinery.

Drive system properties:

1- The voltage controller output voltage is non-sinusoidal voltage having several harmonics, and the main harmonic frequency is 50 Hz, which is the fundamental harmonic and will be taken into account and the rest of the harmonics neglected.

2- The power factor of the system is variable and depends on the firing angle of the thyristor $(\mathbf{Cos\alpha \cong Cos\varphi})$. This is disadvantage of the system, but it is overcome by using means of improving the power factor.

3- The voltage controller output voltage depends on the thyristor firing angle (α) as well as on the load angle (φ_L). The load of the controller is the squirrel cage ac motor, it is resistive inductive load (R-L Load). The angle of the load (φ_L) depends on the value of the slip .

4- The motor electromagnetic torque is calculated from the influence of the first (fundamental) harmonic stator voltage, having the following relationships:

Load angle ; $(\varphi_L = f(S))$

Thyristor firing angle; $(\alpha = f(V_C))$

Where; (V_C) : Control signal

A- load angle relationship:

The load angle with respect to the voltage control output is calculated as shown in Figure (7-5).

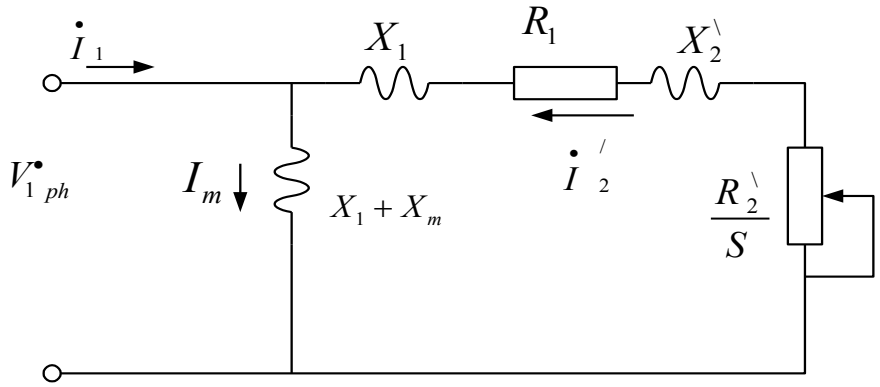


Figure (7-5) Equivalent circuit of an induction motor

Total Equivalent Circuit Impedance: -

$$\mathbf{Z_{eq} = \frac{(Z_1 + Z_2)Z_m}{Z_1 + Z_2 + Z_m} = R_{eq} + jX_{eq}}$$

Where :

$$\mathbf{Z_1 = R_1 + jX_1}$$

$$\mathbf{Z_2 = \frac{R_2'}{S} + jX_2}$$

$$\mathbf{Z_m \cong jX_m}$$

By substituting the values of impedances, we find the relationship of the load angle:

$$\mathbf{\varphi_L = \tan^{-1}\left(\frac{X_{eq}}{R_{eq}}\right)}$$

$$\mathbf{\varphi_L = \tan^{-1}\left(\frac{(R_1 S + R_2')^2 + X_{S.C}(X_m + X_{S.C})S^2}{(R_1 S^2 + R_2' S)X_m}\right)}$$

This load angle is dependent on the slip value $(\varphi_L = f(S))$, as shown in Figure (7-6).

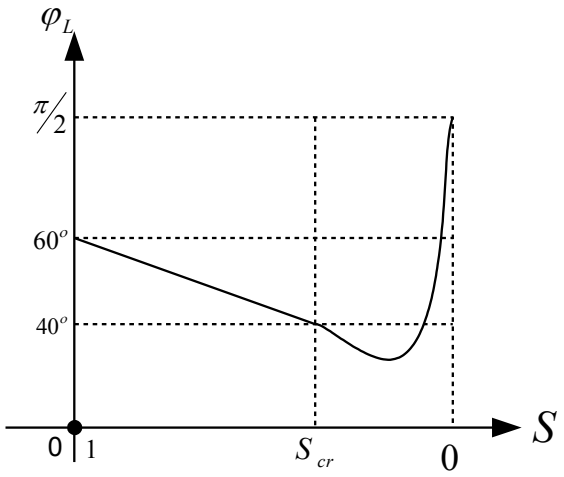


Figure (6-7) The relationship of the load angle

B- The relationship of the thyristor firing angle, as in Figure (7-7).

The control voltage varies with the triggering angle according to the following relationship:

$$\frac{V_C}{V_{Cn}} = 1 - \frac{\alpha}{\alpha_{max}}$$

Where:

α_{max}**: The maximum value of the firing angle .**

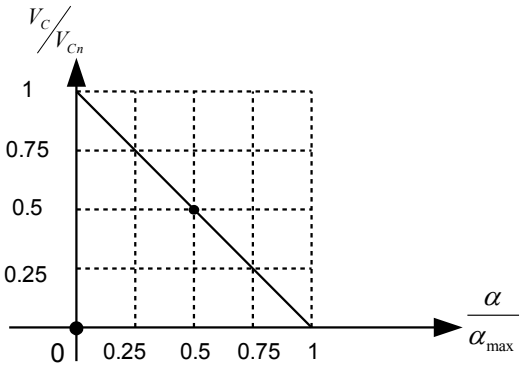


Figure (7-7) The control signal as function of firing angle

Figure (7-8) shows the relationship of the controller output voltage as function of load angle at different values of firing angle.

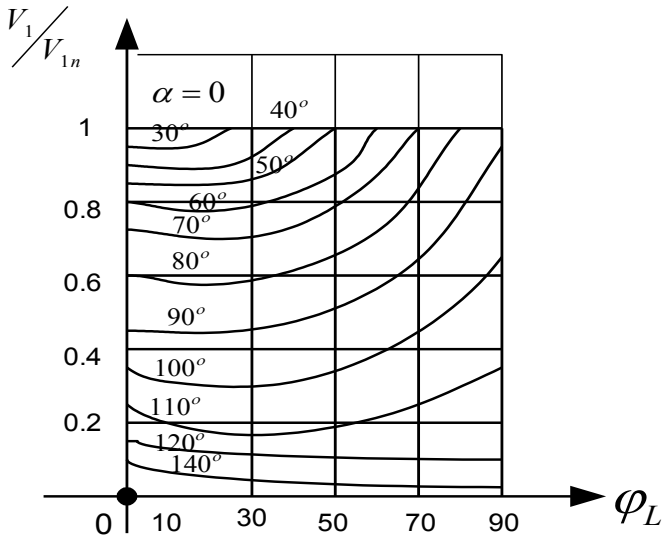


Figure (7-8) The controller output voltage relationship

By knowing the previous relationships, the motor torque-speed characteristics for $(\alpha = \text{Var})$ are drawn using the motor nominal parameters according following steps:

For each value (α), a table is filled in as follows:

1- For the values of slip $(S = 0 \div 1)$.

2- We find the motor torque value $(T_{(N/C)})$ from the natural characteristic of each of the slip values.

3- Determine the load angle for the values of slip from the relationship of load angle.

4- Knowing (α) and the load angle, we find a value $\left(\frac{V_1}{V_{1N}}\right)$.

5- We calculate the motor torque of the regulated characteristic by the following relationship:

$$T = T_{(N/C)}\left(\frac{V_1}{V_{1N}}\right)^2$$

Thus, for a value of (α) we plot the motor torque-speed characteristics $T=f(s)$, according to the table below.

$$\alpha=\alpha_1$$

S	0	0.1	.....	S_{cr}	.....	1
$T_{(N/C)}$	0			T_{max}		$T_{S.C}$
φ_L	90°			40°		60°
$\frac{V_1}{V_{in}}$						
T						

Variable Voltage and Variable Frequency converters

Various types of VVVF converters are used for speed control of induction motors

Frequency converters are divided into two types:

1- Direct (Single- Stage) Frequency Converter.

Figure (7-9) shows the power circuit of direct frequency converter.

Direct frequency converter converts fixed ac nominal voltage and frequency into variable voltage and variable frequency in single stage using half wave controlled rectifier group (R) and inverter group (I) in each phase.

The number of thyristors is 18, it is very complicated to control and synchronize the operation of the thyristors.

This type is rarely used nowadays.

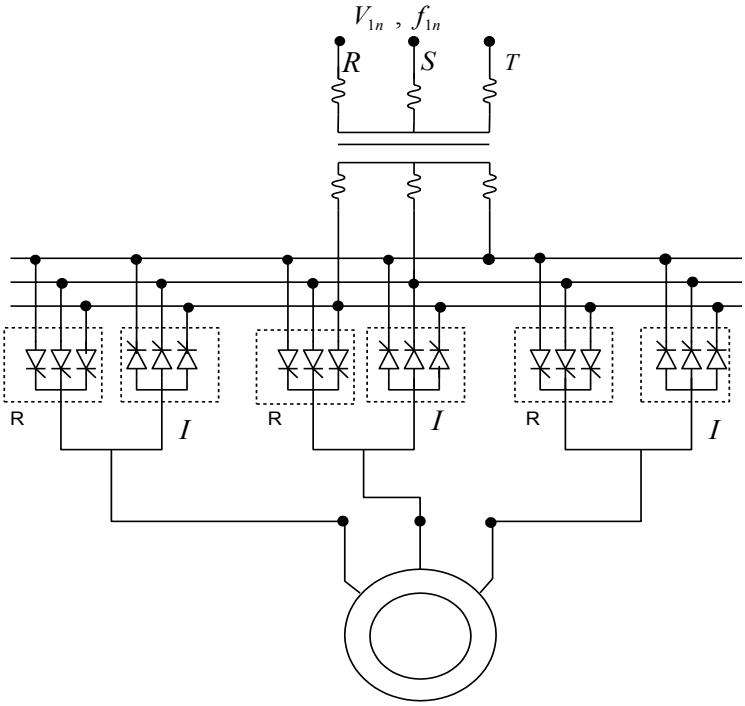
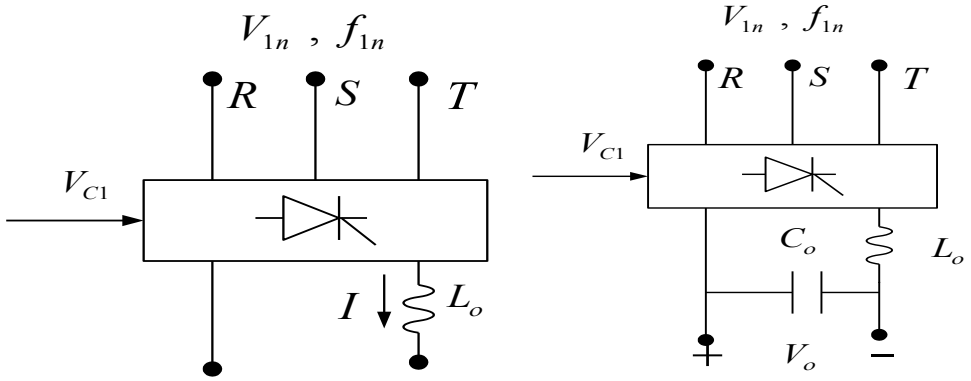


Figure (7-9) The power circuit of direct frequency converter

2- Indirect (Double-Stage) Frequency Converter

The stator voltage and stator frequency are varied through two stages.

The first stage: a rectifier, either controlled or uncontrolled, is used to convert the AC voltage into direct current voltage. The first stage may be in the form of a voltage source as in Figure (7-10-A), or a current source as in Figure (7-10-B).

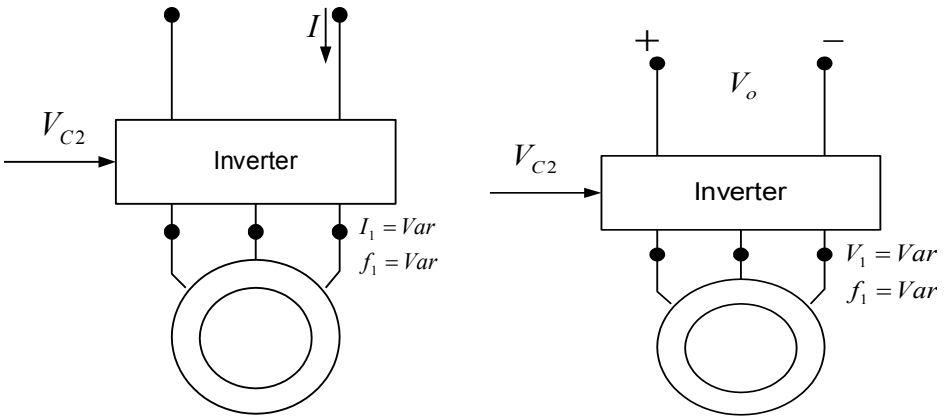


B) current source **A) voltage source**

Figure (7-10) Types of power sources

The second stage: A controlled inverter is used to convert the direct current voltage to AC variable voltage and a variable frequency, and it is called in this Frequency Converter With Voltage Source Inverter.

As shown in Figure (7-11-A), or to convert direct current (current source) into variable current and variable-frequency alternating current, and in this case it is called Frequency Converter With Current Source Inverter, as shown in Figure (7-11-B).



A) Voltage source inverter **B) Current source inverter**

Figure (7-11) types of inverter

The inverter uses the power electronics devices as a thyristor or transistor. Figure (7-12-A) shows the variable voltage and variable frequency converter.

The variable voltage and variable frequency converter consists of the following components:

1- Three-phase full-wave fully controlled rectifier consisting of thyristors $(T_1 - T_6)$ connected to a filter (L_o , C_o) , and a very large value of capacity gives the it a voltage source property. The capacitor works to keep the voltage constant at the output of the controlled rectifier regardless of the change in the load at a given firing angle.

2- Three-phase controlled inverter consisting of thyristors $(T_7 - T_{12})$. It converts direct current power into AC power with variable voltage and variable frequency, where a value (V_1) varies with a variation in value (V_o), and the frequency varies with the variation of the firing angle of the inverter.

3- Diodes $(D_7 - D_{12})$ used to separate the forced commutation circuit $(L-C)$, connected between each of the two phases of the three-phase squirrel cage ac motor circuit.

4- Diodes $(D_1 - D_6)$ used to return the power from the motor to the source capacitor C_o.

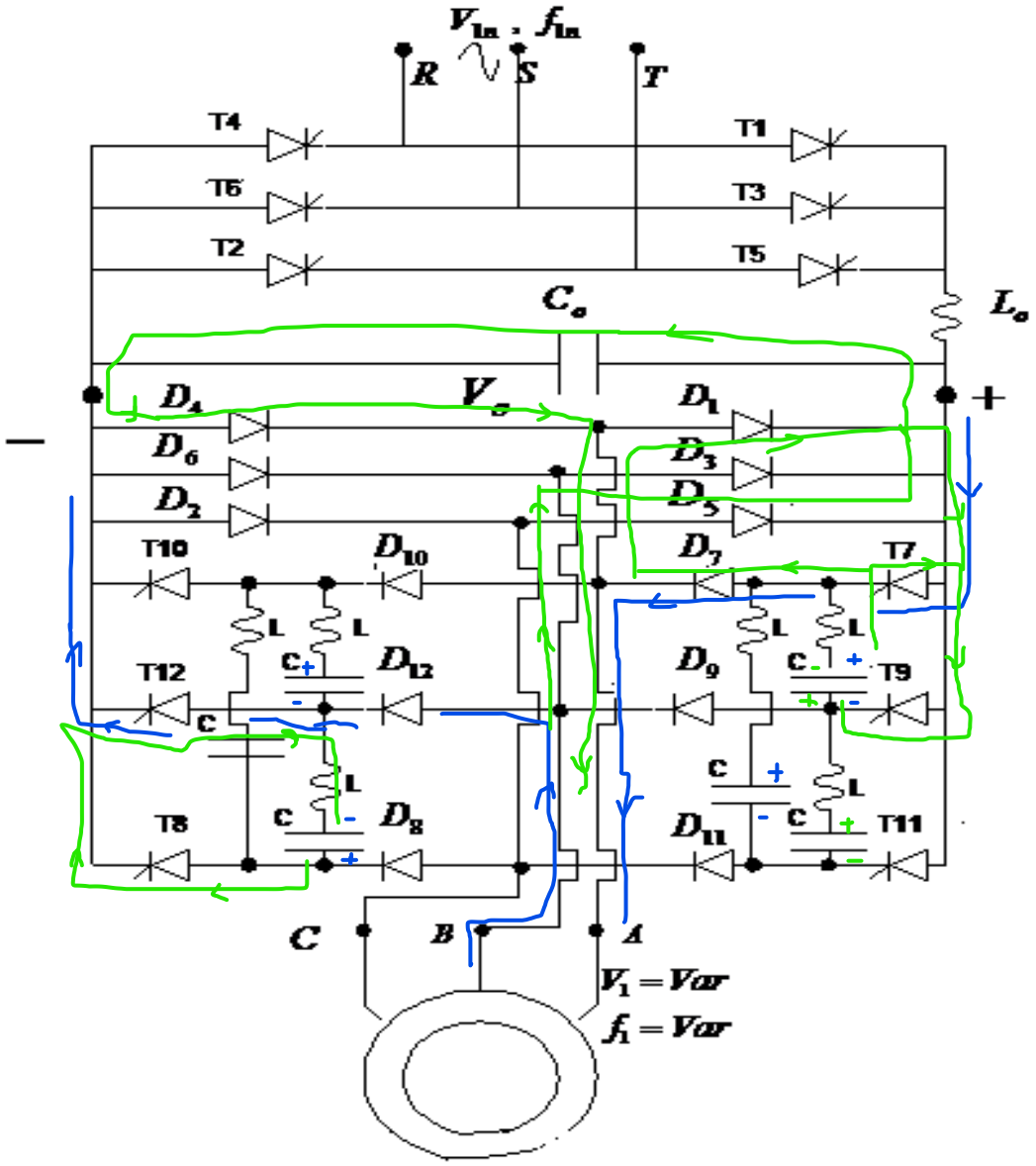


Figure (7-12-A) variable voltage and variable frequency converter

Principle of operation:

Assume that at a certain moment of time the thyristor (T_7) works with (T_{12}) , where the current passes from the positive terminal of the direct current source through the thyristor (T_7), diode (D_7) and phase (A) of the induction motor, then the phase (B), diode (D_{12}) and thyristor (T_{12}), and then the negative terminal of the direct current source.

During this interval, the capacitor connected between (T_9, T_7) and the capacitor connected between (T_{12}, T_8) are charged, so that the upper plate of the capacitor is positive and the lower plate is negative, so that when the thyristors (T_8, T_9) are

turned on, the thyristor (T_{12}, T_7) are forcely commutated when the value of the current passing through them reaches zero.

The two capacitors of commutation circuit are discharged through the next circuit as follows:.

From the positive plate of the capacitor through $(\rightarrow D_7 \rightarrow D_1 \rightarrow T_9 \rightarrow)$ to the negative plate of the capacitor, the capacitor recharged.

In the moment of changing the direction of the voltage due to the turning on of the thyristors (T_8, T_9), the voltage and the current will have two different signals, as shown in Figure (7-12-B). So that the product of them is negative and the power returns to the source through the following circuit:

From phase (B) of the motor through $(\rightarrow D_3 \rightarrow C_o \rightarrow D_4 \rightarrow)$ to phase (A) of the motor.

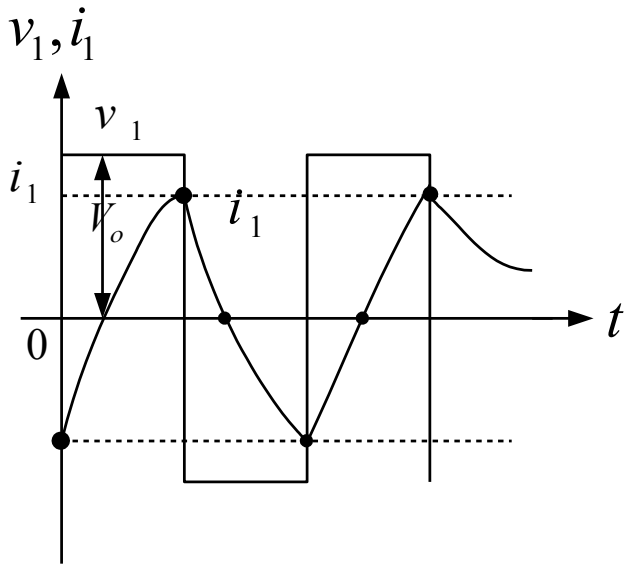


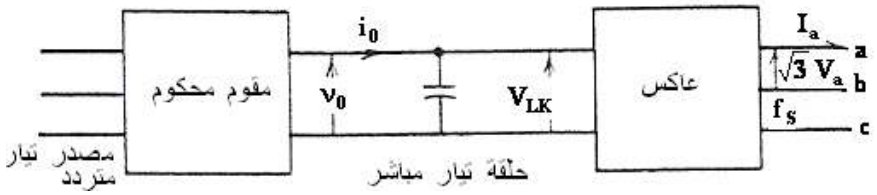
Figure (7-12-B) The ac motor stator voltage and stator current waveforms

Speed Control of AC Motors using Voltage Source Inverter

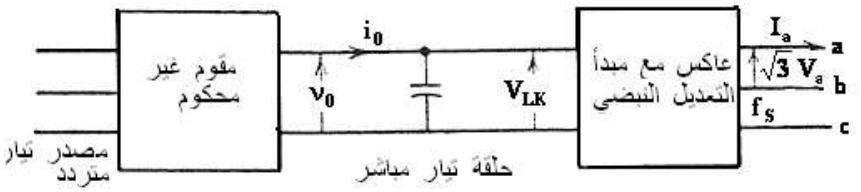
The inverter is fed by direct voltage source and supplies AC motors with variable voltage and variable frequency.

Figure (7-13) shows the types of voltage source inverter with a, where in Figure (7-13-A) the inverter is fed by a Controlled Rectifier that varies the inverter input voltage and at the same time the inverter varies the frequency. The direct current link may consist of a coil that is connected in series to the rectifier output but the most important component is the capacitor, which acts as a filter and keeps the inverter input voltage constant when the load changes. As for Figure

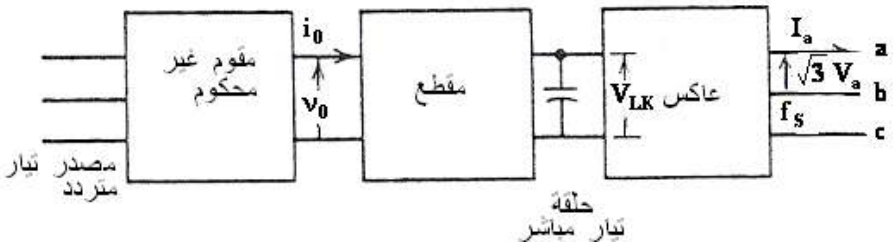
(7-13-B) , the inverter is fed by un-controlled rectifier and for this reason, the inverter changes the voltage and the frequency together according to the principle of pulse modulation (PWM). In Figure (7-13-C) the inverter is fed from the chopper as a variable dc voltage source. Figure (7-13-D) shows VSI using power transistors.



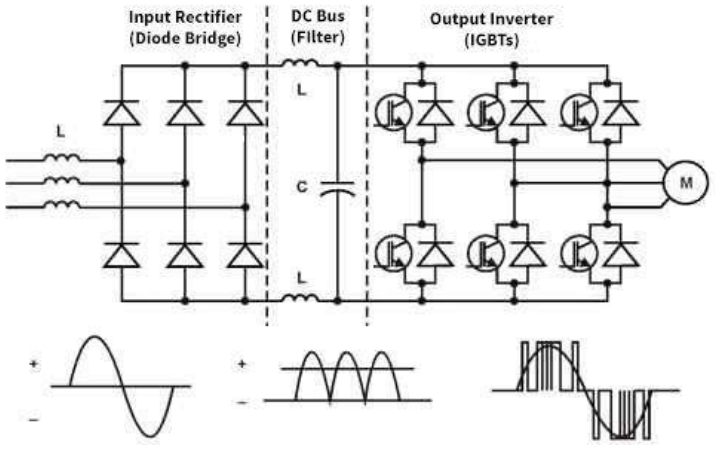
(A)



(B)



(C)



(D)

Figure (7-13) Voltage source inverter configurations

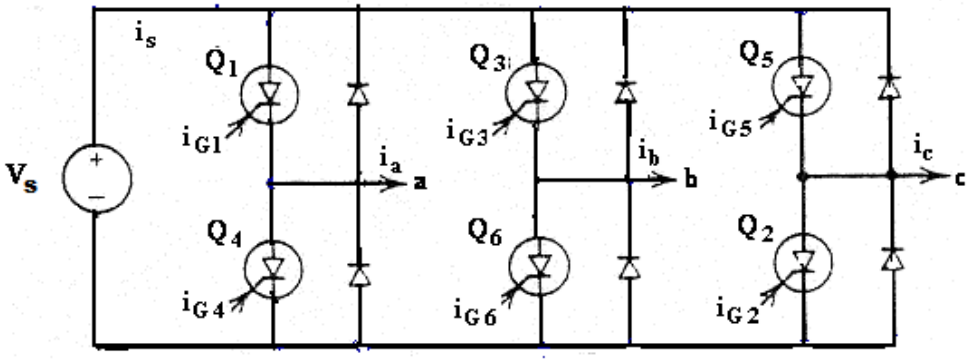
One of the disadvantages of previous inverter circuits is that they do not work to return (regenerating)power to the source, and appropriate circuits must be used when needed to return power to the source.

Voltage Source Inverter Principle of Operation

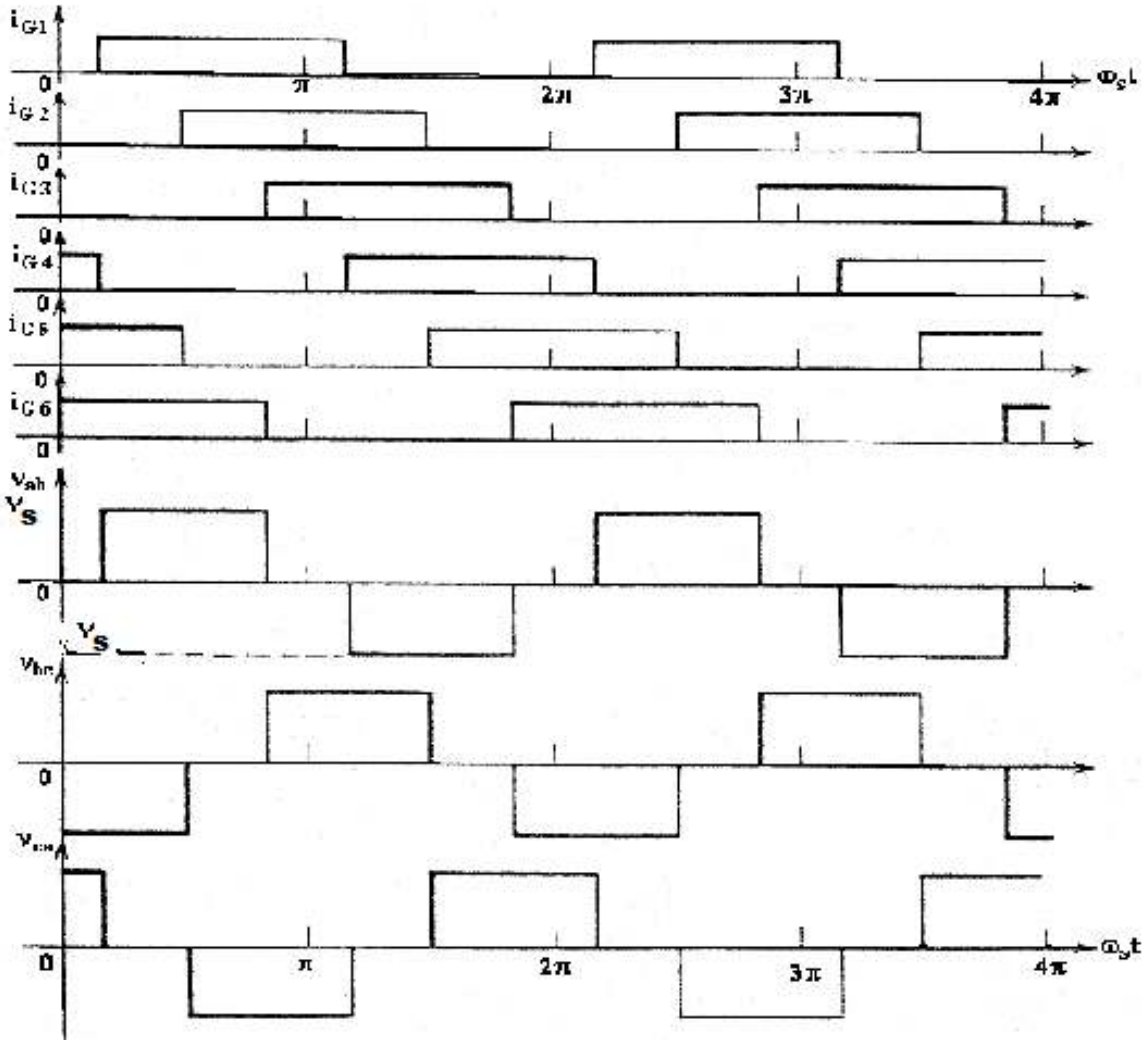
Figure (7-14-A) shows a circuit of a three-phase inverter fed from a voltage source (V_s). The inverter uses thyristors as power elements and they are numbered according to the sequence of gate signals $(i_{G1}, i_{G2}, ..., i_{G6})$ shown in Fig. (7-14-B).

It is worth noting that three thyristors must operate together at any interval of time this mode is called 180 degrees mode of operation. The operation of the thyristor is turned off after operating half a cycle (π) by the method of forced commutation, and it must be taken into account that no short circuit occurs to the voltage source through two thyristors connected in series to the same arm.

When fired, the thyristor forms a short circuit with the diode connected in reverse with it, for example when firing a thyristor (Q_1), point (a) is connected with the positive terminal of the voltage source, and if at the same time the thyristor is fired (Q_2), then point (c) is connected with the negative terminal of the voltage source and thus is formed the line voltage wave between the two points (a, c) and its value (v_{ca}), as shown in Figure (7-14-B). In the same way, the line voltage waves (v_{ab}) and the line voltage (v_{bc}) are formed. It is evident from the above that it is possible to change the voltage range of the AC wave at the output of the inverter by changing the voltage value as in the inverter circuits Figure 7-13 -A) and Figure (7-13-C).



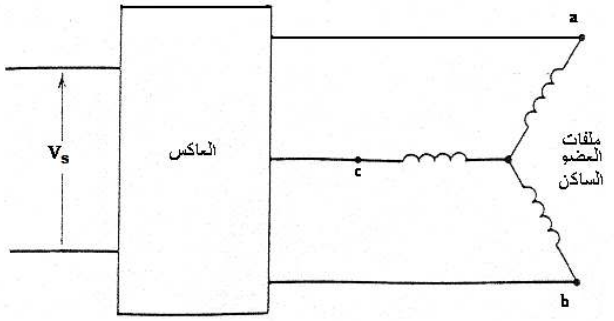
(A) VSI power circuit



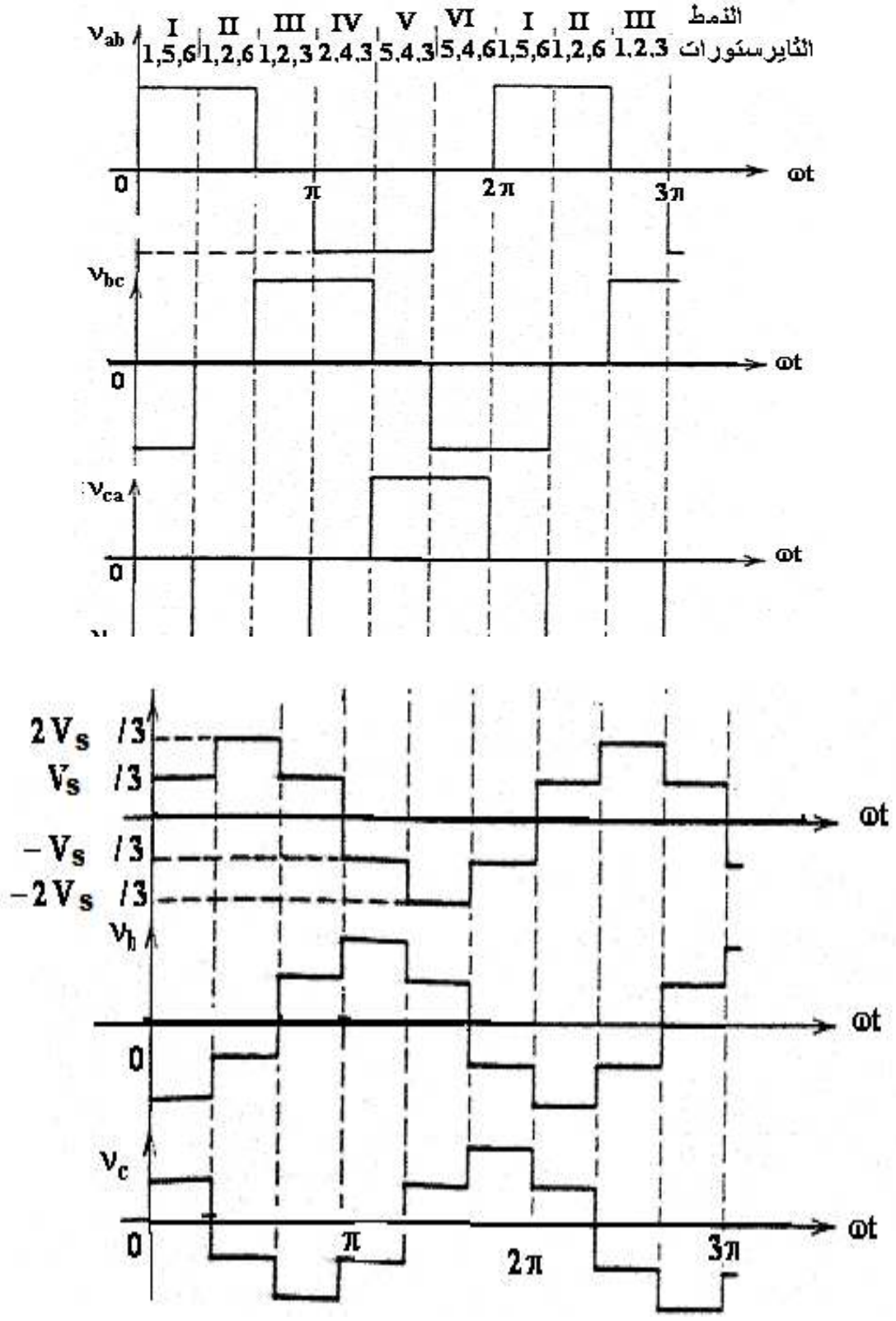
(B) waveform of thyristors conduction sequence and voltages

Figure (7-14) Three-phase voltage inverter

Figure (7-15-A) shows a circuit of a three-phase ac motor connected with the inverter, and Figure (7-15-B) shows the line voltage waves (v_{ab}, v_{bc}, v_{bc}) and the phase voltage waves (v_a, v_b, v_c) and also the sequence of conduction of the thyristors during the six modes of operation of the inverter.



(A) **(a)**



(B)

Figure (7-15) Voltage source inverter- three-phase ac motor power circuit and waveforms

The phase voltage of a three-phase motor according to the Fourier series can be described as follows:

$$v_1 = \frac{8}{3\pi} V_s \left[0.7500\sin\omega t + 0.1500\sin 5\omega t + 0.1071\sin 7\omega t + ... \right] \qquad (7-1)$$

When this voltage is applied to the motor per-phase equivalent circuit as shown Figure (7-16), the armature phase current is:

$$i_1 = i_m + i_2' \qquad (7-2)$$

Due to the large value of the inductance of the magnetic circuit, the harmonic components of the magnetic circuit current and the air gap flux will be few.
The rms value of the phase voltage is:

$$V_{1ph} = 0.4714\, V_s \qquad (7-3)$$

The rms value of the line voltage is:

$$V_{1L} = \sqrt{3} * 0.4714\, V_s = 0.816\, V_s \qquad (7-4)$$

The rms value of the fundamental component of phase voltage is:

$$V_{11} = \frac{8}{3\pi} V_s * \frac{0.7500}{\sqrt{2}} = 0.452\, V_s \qquad (7-5)$$

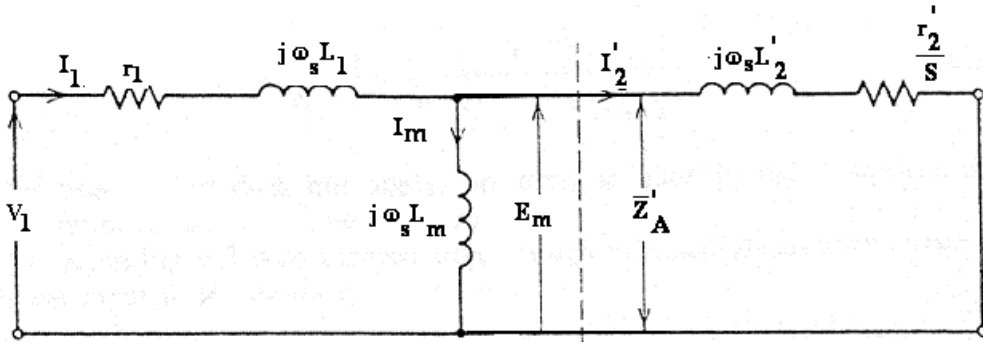


Figure (7-16) Per-phase motor equivalent circuit

The previous method of inverter control is called (180 degrees inverter control mode), in which each switch operates by displaying a continuous pulse of 180 degrees and at any point in time three switches work together.

There is another way to control the inverter called (120 degrees inverter control mode), whereby each switch operates by displaying a continuous pulse of 120 degrees and at any point in time two keys work together.

In this case the rms value of the phase voltage is:

$$V_{1ph} = 0.409\, V_s \qquad (7-6)$$

The rms value of the line voltage is:

$$V_{1L} = \sqrt{3} * 0.409 V_s = 0.707 V_S \qquad (7-7)$$

Comparison between 120 degrees and 180 degrees inverter control modes

NN	Parameter	120 degrees Inverter control mode	120 degrees Inverter control mode
1	Switch conduction interval	2x60 degrees	3x60 degrees
2	Switches sequence of operation	12,23,34,45,56,61	123,234,345,456,561,612
3	Output phase voltage wave form	120 degree quasi-square segments with dead zone symmetrical around time axis with 120 degrees phase shift	Six stepped segments symmetrical around time axis with 120 degrees phase shift
4	Output line voltage wave form	Six stepped segments symmetrical around time axis with 120 degrees phase shift	120 degree quasi-square segments with dead zone symmetrical around time axis with 120 degrees phase shift
5	Output voltage harmonics	Only odd harmonic present	Only odd harmonic present
6	Phase rms voltage	0.409 V_{dc}	0.47 V_{dc}
7	Line rms voltage	0.707 V_{dc}	0.816 V_{dc}
8			

The principle of pulse-width modulation

In the previous case, the principle of pulse modulation was used to give one continuous pulse during the operation of the switch, and as shown in Figure (7-17), it is possible to use the principle of pulse modulation by several pulses, so that in each half wave several separate pulses are given and therefore in each cycle there are intervals of voltage where three-phase line voltage being zero. This state is obtained by firing a thyristors (Q_1, Q_3, Q_5) or firing a thyristors (Q_2, Q_4, Q_6) at the same time, so that the points (a, b, c) have the same voltage. It can be seen from Figure (7-17) that each thyristor operates three times in one cycle instead of once. In practice, more than two separate pulses are used in each half-wave, which may be asymmetric. This causes low-frequency harmonics to appear. In practical applications there is a limitation of the number of pulses in the half-wave for the following reasons:

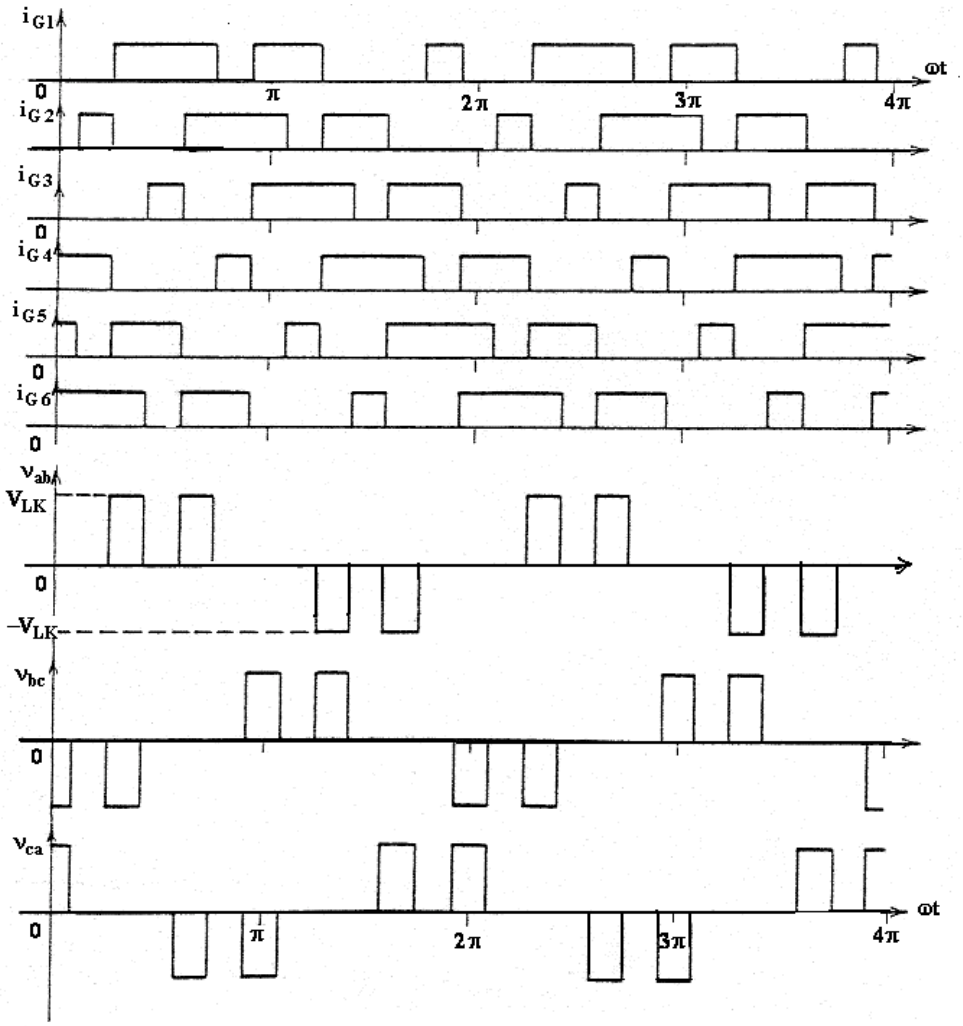


Figure (7-17) Pulse Width Modulated voltage source inverter

1- Thyristor switching process requires a specific time, which requires determination of the pulse frequency.

2- The process of switching the thyristor is accompanied by power losses. Increasing the frequency of the pulses causes more losses and reduces the efficiency of the inverter.

The inverter output voltage waveforms shown in Figure (7-15) and Figure (7-17) cause vibrations in the motor shaft especially at low speeds and in starting conditions. This problem can be overcome by using the principle of sinusoidal pulse modulation.

Principle of Sinusoidal Pulse-Width Modulation

Figure (7-18-A) shows the process of generating thyristor firing pulses by comparing the sinusoidal reference signal for each phase with the sawtooth signal so that the required thyristors are connected and disconnected as shown in Figure (7-14-A), which leads to the appearance of the two phase voltages (V_a, V_b) as in the figure (7-18-B).

We obtain the line voltage (V_{ab}) as a resultant of the voltages of the two phases (V_a, V_b), as in Figure (7-18-C).

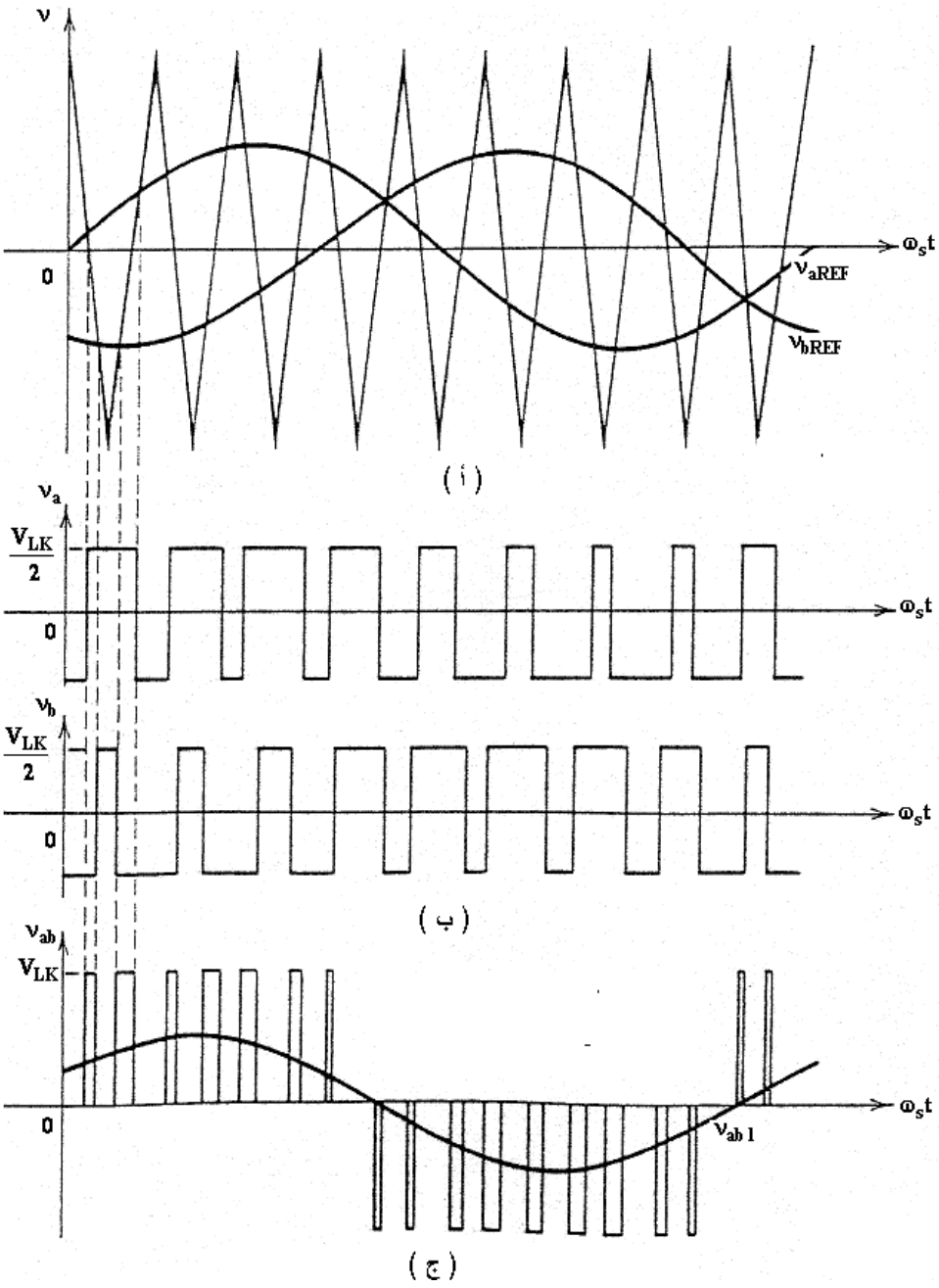


Figure (7-18) Sinusoidal pulse modulation width modulated voltage source inverter

Three-Phase AC Motor and Voltage Source Inverter Ratings

The maximum apparent power at the output of the inverter is:

$$S_I = 3V_{1max} I_{1max} \quad VA \qquad (7-8)$$

The motor per- phase equivalent circuit Figure (7-18) is used, where the stator's impedance is neglected to extract the mathematical relationships for the ideal match between the motor power and the inverter power.

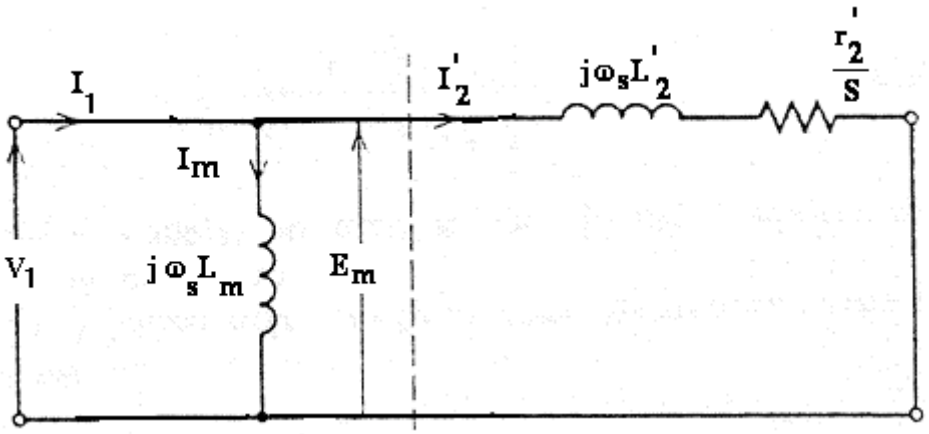


Figure (7-18) Motor per-phase equivalent circuit

We write the motor torque relationship as follows:

$$T = \frac{3}{\omega_s}\frac{P}{2}\frac{r_2'}{S}\frac{V_1^2}{(\frac{r_2'}{S})^2 + (\omega_s L_2')^2} \qquad (7-9)$$

The previous relationship was written considering that:

$$r_1 = 0 \, , x_1 = 0 \, ;$$

$$\omega_0 = \frac{2\pi f_1}{(P/2)} = \frac{\omega_s}{(P/2)}$$

Where:

ω_0 **: Stator field angular speed,** $\left[\frac{rad}{s}\right]$

ω_s **:Stator angular frequency,**

P :The stator number of poles

By deriving the relationship (7-10) and equating the result to zero, we find the critical slip relationship, which is:

$$S_{cr} = \pm \frac{r_2'}{\omega_s L_2'} \qquad (7-11)$$

Substituting the relationship (7-11) in the relationship (7-10), we find the relationship of motor maximum torque as follows:

$$T_{max} = \frac{3PV_1^2}{4\omega_s^2 L_2'} \qquad (7-12)$$

Consider with a slight error that the stator angular frequency is:

$$\omega_S = \frac{P}{2}\omega_0 \qquad (7-13)$$

Substituting the relationship (7-13) in the relationship (12-7) we write the motor maximum torque relationship as follows:

$$T_{max} = \frac{3V_1^2}{P\omega_O^2 L_2'} \qquad (7-14)$$

This means that the maximum motor torque is inversely proportional to the square of the stator field angular speed in case of speed control with fixed stator voltage as follows:

$$T_{max} \propto \frac{1}{\omega_O^2} \qquad (7-15)$$

At light loads the process of motor speed control is carried out with the power constant above the natural characteristic with the stator voltage constant at the nominal value according to the following relationship:

$$T \propto \frac{1}{\omega_0} \qquad (7-16)$$

Figure (7-19) shows the torque-speed characteristics when controlling the motor speed with constant load torque and constant load power. Also shown on the figure are the power stability curve $(T\omega_0 = K_1)$ and the torque curve $(T_{max}\ \omega_0^2 = K_2)$ that the motor can provide without failure.

The motor can operate at constant power at the rated voltage and current up to the point of intersection of the two curves, the shaded area between the two curves indicates the extent to which the motor cannot provide the required acceleration, because the maximum motor torque is less than the torque required to operate at constant power and therefore the motor current must be reduced. To ensure acceptable acceleration conditions, the power of the inverter must be greater than that of the motor.

As shown in Figure (7-19), a motor with maximum torque equal to twice the rated torque can operate at constant power only up to a maximum speed equal to twice the rated speed. (λn=2)

We consider that the power stability curve $(T\omega_0 = K_1)$ and the torque curve that the motor can provide without collapsing $(T_{max}\ \omega_0^2 = K_2)$ intersect at a maximum speed of its magnitude (ω_{max}), and accordingly the following relationship can be written:

$$\frac{K_1}{\omega_{max}} = \frac{K_2}{\omega_{max}^2} \qquad (7-17)$$

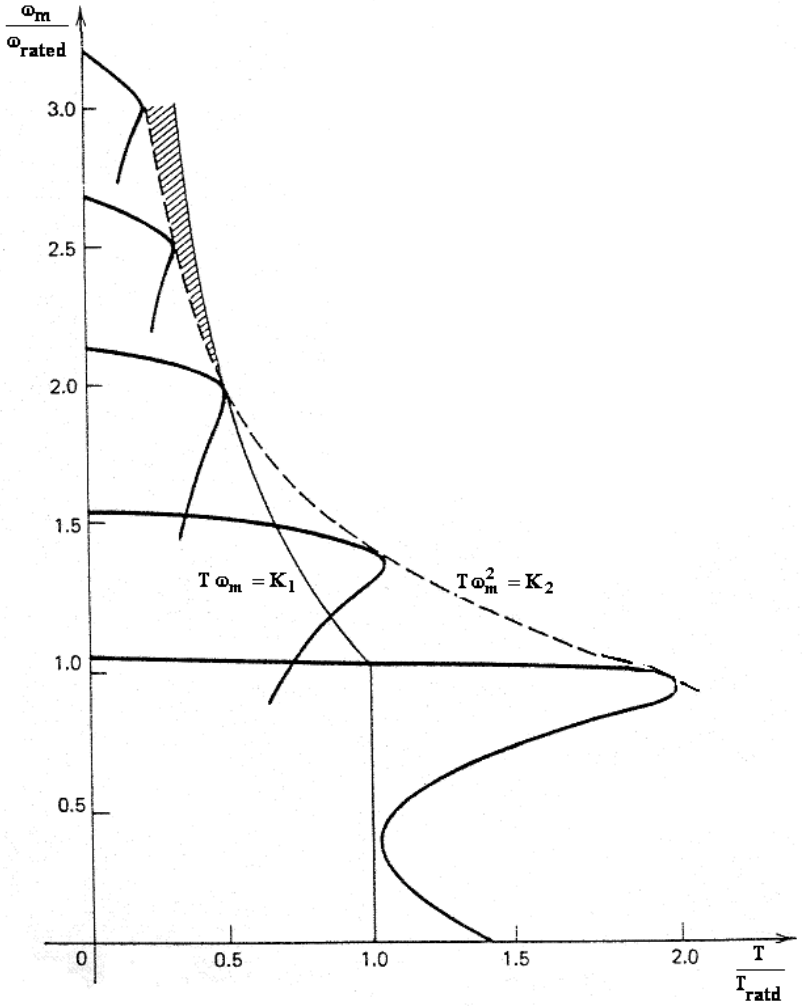


Figure (7-19) Motor torque-speed characteristics for constant load torque and constant load power

When the motor is running at the rated load and rated speed, the motor power is:

$$T_N \omega_N = K_1 \qquad (7-18)$$

By substituting the relationship (7-18) into the relationship (7-17), we find:

$$K_2 = T_N \omega_N \omega_{max} \qquad (7-19)$$

AS, $\left(T_{max}\, \omega_N^2 = K_2\right)$**, then:**

$$T_{max} = \frac{T_N \omega_{max}}{\omega_N} \qquad (7-20)$$

$$\frac{T_{max}}{T_N} = \frac{\omega_{max}}{\omega_N} \qquad (7-21)$$

The process of starting in industrial machinery is done by increasing the voltage and the frequency together gradually and at nominal stator current i.e. with the constant nominal load torque, and after nominal stator voltage is reached, the motor acceleration process above natural characteristic is carried out by increasing the frequency only and under the influence of weakening the motor magnetic flux with constant load power due to the limited power of the source.

Figure (7-20) shows the relationship of the change of voltage and stator current with the change of the motor angular speed, where it can be seen that the motor operates at a constant current value up to twice the prescribed speed, and then the motor current begins to decrease because the maximum torque of the motor becomes less than the torque required to work at constant power .

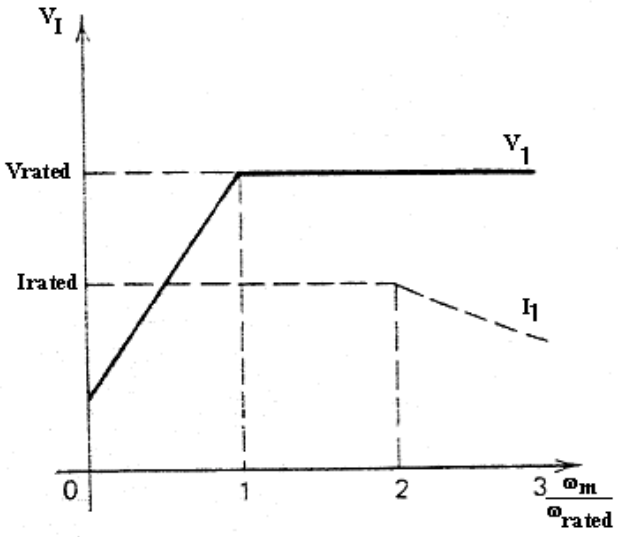


Figure (7-20) The relationship of stator voltage and stator current change with change of motor speed

From the relationship (7-21), we find that for the motor to operate at a constant current value up to three times the specified speed, the value of the motor maximum torque must be equal to three times the rated torque. This means that a more powerful motor should be used with more data than the inverter.

To overcome this problem, it is possible to regulate the speed in this range by using a motor with a lower power by increasing the stator voltage over the rated value.

In the following, we will find the relationship of the voltage change with the change in speed over twice the rated speed:

We write the relationship (7-14) for speed $\left(\omega_{01}\right)$ and voltage $\left(V_{a1}\right)$, and we find:

$$T_{max\,1} = \frac{3V_{a1}^2}{P\omega_{01}^2 L'_2} \qquad (7-22)$$

In the same way at speed $\left(\omega_{02}\right)$ and voltage $\left(V_{a2}\right)$, we find:

$$T_{max\,2} = \frac{3V_{a2}^2}{P\omega_{02}^2 L'_2} \qquad (7-23)$$

From the two relationships (7-22) and (7-23) and considering that the motor is running at constant power, that is:

$$T_{max1}\omega_{01} = T_{max2}\omega_{02} \qquad (7-24)$$

so,T_max1 ω_01=(3V_a1^2)/(Pω_01 L_2^')
=T_max2 ω_02=(3V_a2^2)/(Pω_02 L_2^')

we find that :

$$\frac{V_{a1}}{V_{a2}} = \sqrt{\frac{\omega_{01}}{\omega_{02}}} \qquad (7-25)$$

If we know that the speed at the intersection of the two curves is $\left(\omega_{01}\right)$ and the motor voltage value $\left(V_{a1} = V_{1N}\right)$, then the voltage value at the higher speed $\left(\omega_{02}\right)$ is:

$$V_{a2} = V_{1N}\sqrt{\frac{\omega_{02}}{\omega_{01}}} \qquad (7-26)$$

From the relation (7-26) we write the maximum value of the stator voltage $\left(V_{a\,max}\right)$ required at the maximum speed $\left(\omega_{m1}\right)$:

$$V_{a\,max} = V_{1N}\sqrt{\frac{\omega_{max}}{\omega_{01}}} \qquad (7-27)$$

Since the power of the inverter output in this range of speed regulation is constant, the stator current relationship can be written as follows:

$$I_{a2} = I_{1N}\sqrt{\frac{\omega_{01}}{\omega_{02}}} \qquad (7-28)$$

Figure (7-21) shows the relationship of the change in voltage and current at high speeds.

By changing the stator voltage at high speeds, we obtain acceleration and speed regulation processes at constant power and by using a lower power motor. But the power of the inverter increases according to the relationship (7-8) because the power devices have to withstand the current and supply the motor with the required voltage.

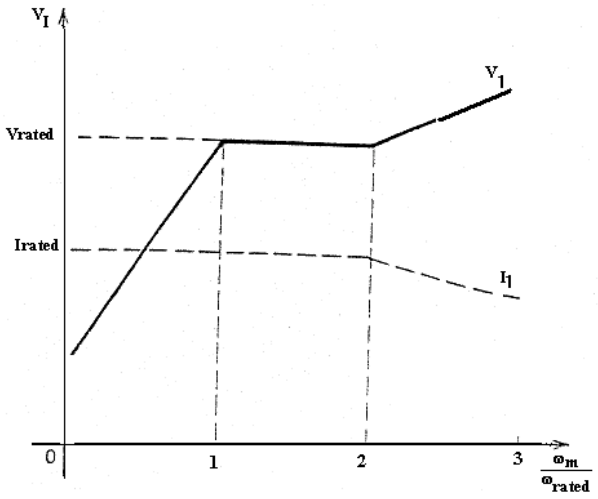


Figure (21-7) the relationship of voltage and current change at high speeds

Usable Inverter Power

The usable inverter power (S_U) is part of the apparent inverter power and depends on the switching frequency of the thyristor, and it is expressed in the following relationship:

$$S_u = K_P V_{amax} I_{amax} \quad VA \qquad (7-29)$$

Where :

K_P :Usable inverter power factor.

Figure (7-22) shows the relationship of the change of the usable power factor of the inverter with the switching frequency of the thyristor, as it is found that the higher the frequency value, the less the usable power of the inverter, due to the increase in switching losses on the power devices and the decrease in the inverter output voltage due to the increase in dead time intervals, between the processes of conducting and disconnecting the thyristors.

The switching frequency of the thyristors can be reduced by regulating the inverter voltage using controlled rectifier or chopper, but this will increase the dimensions and costs of the voltage regulator as a whole.

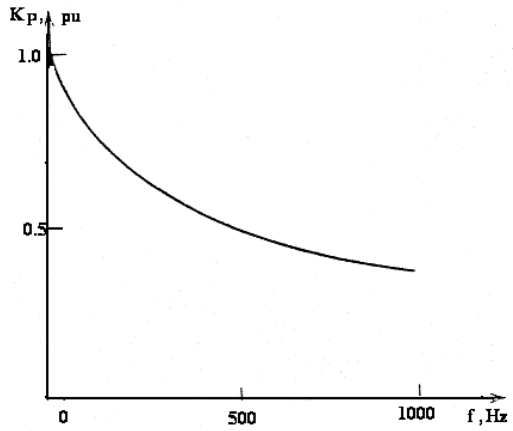


Figure (7-22) The relationship of the change of the usable power factor of the inverter with the switching frequency of the thyristor

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